INCH-POUND
MIL-DTL-52869C
4 August 1997
SUPERSEDING
MIL-S-52869B
22 February 1980

DETAIL SPECIFICATION

SENSOR GROUP, PASSIVE ULTRASONIC: RECEIVER R-1860/FSS-9(V) AND PROCESSOR MX-9443/FSS-9(V)

Inactive for new design after 4 August 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a receiver and a processor which combine to form a passive ultrasonic sensor, designed to detect the sounds of forced entry into a protected space.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Standards and handbooks. The following standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-889	-	Dissimilar Metals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army CECOM, ATTN: AMSEL-LC-IEW-D-ED, 10115 Gridley Rd. Suite 228, Ft Belvoir VA 22060-5849 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6350
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-454	-	General Guidelines for Electronic Equipment.
MIL-HDBK-781	-	Reliability Test Methods, Plans and Environments for Engineering Development, Qualification and Production.

(Unless otherwise indicated, copies of the above standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia PA 19111-5094.)

2.2.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

TA13220E3101	-	Processor, Passive Signal Ultrasonic, MX-9443/FSS-9(V).
TA13220E3111	-	Circuit Card Assembly, Processor.
TA13220E3151	-	Receiver, Passive Signal Ultrasonic, RT-1860/FSS-9(V).
TA13220E3163	-	Circuit Card Assembly, Receiver.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from: USA CECOM, IEWS Directorate, ATTN: AMSEL-LC-IEW-D-CE, 10115 Gridley Rd., Suite 228, Ft Belvoir VA 22060-5849).

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ASQC-Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes.
ISO 10012-1	-	Quality Assurance Requirements For Measuring Equipment - Part 1: Metrological Confirmation System For Measuring Equipment.

(Application for copies should be addressed to: The American National Standards Institute (ANSI), 11 West 42nd Street, New York, NY 10036).

AMERICAN WELDING SOCIETY (AWS)

D9.1 - Welding of Sheet Metal.

(Application for copies should be addressed to: American Welding Society, 550 NW LeJeune Road, PO Box 351040, Miami FL 33135.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Description. The Passive Ultrasonic Sensor Group (PUS) shall consist of one processor, passive signal ultrasonic, MX-9443/FSS-9(V), (hereinafter called "processor"), combined with a minimum of one to a maximum of twenty receivers, passive signal ultrasonic, R-1860/FSS-9(V) (hereinafter called "receivers"). The processor shall be in accordance with TA13220E3101 and as specified herein. The receiver shall be in accordance with TA13220E3151 and as specified herein.

3.1.1 Major components. The major components of the PUS shall be as follows:

a. Processor, Passive Signal Ultrasonic, TA13220E3101.
b. Circuit Card Assembly (CCA), Processor, TA13220E3111.
c. Receiver, Passive Signal Ultrasonic, TA13220E3151.
d. Circuit Card Assembly, Receiver, TA13220E3163.

When specified (see 6.2), any of the major components listed above shall be furnished separately.

3.1.2 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly and operation of the PUS. Any data (for example, shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The PUS shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the PUS may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the PUS.

3.4 Environmental conditions. The PUS and major components when furnished separately shall withstand individual environmental conditions specified herein, or combinations thereof, without damage (see 6.3.3), performance degradation, or malfunction when tested as specified in 4.4.2.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

3.4.1 Operational.

a. Temperature range of -20 to +150 °F.
b. Humidity up to 95 percent relative.

3.4.2 Non-operational condition and storage.

a. Temperature range of -30 to +165 °F.
b. Humidity up to 95 percent relative.
c. Shock: bench handling, equipment not operating.

3.5 Detailed technical requirements. The PUS and major components when furnished separately shall perform as specified herein under any of the operating test conditions specified herein.

3.5.1 Terminal board connections.

3.5.1.1 Receiver terminal board. The receiver terminal board shall be marked with identifying terminal numbers as follows:

Terminal Board 1 (TB1)	Function
1	receiver output signal
2	receiver output signal return
3	spare
4	spare
5	tamper signal
6	tamper signal
7	spare
8	spare
9	enclosure ground
10	spare

3.5.1.2 Processor terminal boards. The processor terminal boards shall be marked with identifying terminal numbers as follows:

Terminal Board 1 (TB1)	Function
1	alarm output
2	alarm output return
3	spare
4	spare
5	tamper signal
6	tamper signal
7	+20 volts direct current (Vdc) input
8	20 Vdc return
9	enclosure ground
10	spare

On TB1, removable jumpers shall be provided across terminals 2 and 3, 3 and 4, and 4 and 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

Terminal Board 2 (TB2)	Function
1	receiver input signal
2	receiver input signal return
3	spare
4	spare
5	spare
6	spare
7	synchronization (sync) signal input
8	sync signal return (signal ground)
9	enclosure ground
10	spare

Terminal Board 3 (TB3)	Function
1	input to notch rejection filter
2	input signal from receiver
3	output to signal processor circuitry
4	notch rejection filter output

3.5.2 Functional characteristics.

3.5.2.1 Power. The PUS shall operate from power supplied to the processor from an external 20 Vdc regulated power supply. Reverse polarity connections to the processor shall not damage the PUS. A PUS consisting of a processor and any number of receivers from one to 20 shall operate as specified herein on regulated voltage between 18 and 22 Vdc with a current drain not exceeding 60 miliamperes (mA).

3.5.2.2 Performance. The PUS shall be sensitive to all ultrasonic energy within the frequency range of 20 to 30 kilohertz (kHz) that is not rejected by the notch (band-reject) filter incorporated in the processor and that is more than 50 decibels (dB) above the reference sound pressure level (RSPL) of 0.0002 microbars at the receiver microphone. The PUS shall perform as specified in 3.5.2.2.1 through 3.5.2.2.4 under the following conditions, except as specified:

a. ultrasonic frequency: 20 to 30 kHz.
b. ultrasonic energy level at receiver microphone: 60 dB above the RSPL.
c. processor sensitivity control: set at maximum.
d. receiver gain control: set at mid-range.
e. number of receivers: any number from one to 20.
f. cable between processor and each receiver: up to 500 feet of cable, RG 58/U or equivalent.
g. band-reject filter: bypassed, except in the case of 3.5.2.2.4.

3.5.2.2.1 Alarm criteria. The PUS shall be in the alarm state after receipt of four energy impulses having a minimum width of 40 milliseconds (ms) within a 15 second period, or receipt of more than 4.5 seconds of continuous ultrasonic energy. Any two or more consecutive impulses occurring within less than 1.5 seconds shall be processed as a single impulse.

3.5.2.2.2 Bandpass characteristics. With the processor sensitivity control set to position 3 and an ultrasonic source set so as to produce a 25.5 kHz signal at an amplitude of 0.5 volts root-mean-square (Vrms) at test point 2 (TP2) in the processor, the response at TP2 shall not differ by more than 4 dB above or 8 dB below the average in-band response over the frequency range from 20 to 30 kHz.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

3.5.2.2.3 Sensitivity control range. The processor sensitivity control shall provide a sensitivity adjustment of not less than 24 dB.

3.5.2.2.4 Notch filter characteristics. The notch filter shall provide 36 dB attenuation of signals having frequencies that are within one percent of a sync frequency within the range 20 to 30 kHz applied to TB2-7 in the processor. The attenuation of signals having frequencies that are not within 5 percent of the sync frequency shall be less than 3 dB.

3.5.2.3 Detailed circuit requirements.

3.5.2.3.1 Gain control range. The receiver gain control shall provide a gain adjustment of at least 20 dB.

3.5.2.3.2 Alarm duration. When the alarm criteria are met (see 3.5.2.2.1), the PUS shall go into the alarm state for not less than 0.5 seconds nor more than 1.5 seconds.

3.5.2.3.3 Alarm output. The resistance across the processor alarm output terminals (TB1-1 and TB1-2) shall be as follows:

a. Non-alarm state: less than 150 ohms.
b. Alarm state: more than 100 kilohms (k ohms).

The alarm output terminals shall be capable of accepting up to 10 mA dc in the non-alarm state and up to 30 Vdc in the alarm state.

3.5.2.3.4 Power interruption. When power to the PUS is interrupted, or the ground wire is disconnected, the PUS shall be in the alarm state.

3.5.2.4 Tamper protection. The resistance between terminals 5 and 6 of the receiver and processor TB1 shall represent a tamper alarm condition by exceeding 100k ohms, or a non-tamper condition by being less than 10 ohms. The tamper alarm condition shall be presented whenever the cover is opened from 0.062 to 0.250 inch (flanges engaged) and shall continue until the cover is closed or the tamper switch actuator rod is pulled out. The switch shall reset by closing the cover.

3.5.2.5 Receiver and receiver CCA. The receiver and receiver CCA shall be in accordance with top assemblies TA13220E3151 and TA13220E3163 respectively and have the characteristics stated below. Unless otherwise specified, the microphone shall be removed, all input and output signals shall be sinusoidal, the sensitivity control shall be set to position 5, and 5 ±0.5 Vdc shall be present across TB1-1 (positive) and TB1-2 (ground) when performing the measurements.

a. The input current shall be less than 1 mA dc.

b. With a 25 kHz input signal applied across terminals 3 and 1 for the microphone (that is, the solder pads on the CCA from which microphone leads 3 and 1 were removed) at an amplitude adjusted to produce an output of 50 millivolts (mV) peak-to-peak (P-P) across TB1-1 and TB1-2, the output shall not vary more than ±5 mV P-P when the frequency is varied from 20 kHz to 30 kHz.

c. With a 22 kHz signal applied to terminals 3 and 1 for the microphone, the signal level at TB1-1 shall decrease by at least 20 dB when the receiver gain control is decreased from position 10 to position 1.

d. With the microphone installed on the receiver CCA, and an ultrasonic transmit transducer on-axis one foot from the microphone, providing a sound pressure level (SPL) at the microphone 75 dB above RSPL (0.0002 microbars) at a frequency of 25 kHz, the signal at TB1-1 shall be 20 ±8 mV root-mean-square (rms) when tested at room temperature (75 ±5 °F).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

e. The voltage across TB1-1 and TB1-2 shall be 5 ±0.5 Vdc with a regulated voltage of 20 ±0.3 Vdc applied through a 24k ohm resistor to TB1-1 with TB2 grounded (see figure 1 of Appendix A) and with the microphone removed and when the 20 ±0.5 Vdc is applied through an 18k ohm resistor with the microphone installed.

3.5.2.6 Processor and processor CCA. The processor and processor CCA shall be in accordance with top assemblies TA13220E3101 and TA13220E3111 respectively and have the following characteristics when 20 ±1 Vdc is applied across TB1-7 (positive) and TB1-8 (negative), and the sensitivity control is set to maximum, except as specified:

a. The input current shall not exceed 35 mA. When input polarity is reversed (TB1-8 connected to positive power supply terminal), the input current shall not exceed 1 mA. When the polarity is returned to its original condition, the input current shall not exceed 35 mA. The voltage at TP5 shall be 12 ±0.6 Vdc. When the input voltage is varied from 18 to 22 V, the voltage at TP5 shall not vary more than ±100 mV.

b. With the sensitivity control set to maximum, a 25.5 kHz signal of amplitude -75 ±3 decibels referred to one milliwatt (dBm) applied through a 0.1 microfarad coupling capacitor to TB2-1 and TB2-2 for 5 seconds shall cause the processor to indicate the alarm state (more than 100k ohms across TB1-1 and TB1-2). When a 40-millisecond (ms) burst of a -75 ±3 dBm test signal is applied every 4 seconds, the processor shall indicate the alarm state on the fourth burst and remain in alarm for 0.5 to 1.5 seconds.

c. The response measured at TP2 shall differ not more than ±3 dB from the average in-band response between 20 and 30 kHz. The response shall be down at least 35 dB at 10 kHz and 20 dB at 40 kHz from the average in-band response between 20 and 30 kHz.

d. The sensitivity control shall provide a sensitivity adjustment of at least 24 dB.

e. The notch filter shall provide 36 dB attenuation of signal frequencies that are within one percent of a sync frequency within the range of 20 to 30 kHz applied to TB2-7. Attenuation of signal frequencies that are not within 5 percent of the sync frequency shall be less than 3 dB.

f. When the alarm criteria in b. are met, the alarm circuit between TB1-1 and TB1—2 shall remain open for not less than 0.5 seconds and not more than 1.5 seconds.

g. The resistance across the alarm output terminals TB1-1 and TB1-2 shall be less than 150 ohms in the non-alarm state and more than 100k ohms in the alarm state.

h. The resistance across tamper switch terminals TB1-5 and TB1-6 shall be less than 10 ohms when the tamper switch is depressed and more than 100k ohms when the tamper switch is released.

3.6 Electromagnetic susceptibility. The electromagnetic susceptibility characteristics of the PUS shall conform to MIL-STD-461, methods RS103 and CS101, as modified in 3.6.1 and 3.6.2.

3.6.1 Radiated. Radiated susceptibility shall be conducted in an electric field of 3 volts per meter (V/m) for both steady-state and keyed conditions of the EMI field generating equipment. This field applies to frequencies up to 500 megahertz (MHz). Above 500 MHz, the electric field shall be 1 V/m. Maximum applied frequency shall be 10 gigahertz (GHz).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

3.6.2 Conducted. Conducted interference signals shall be applied to both power and signal lines and shall be continuous wave, 30 Hz to 50 kHz.

3.7 Reliability. When tested as specified in 4.4.2.7, the PUS shall have a design mean-time-between-failure (MTBF) of 10,000 hours or greater.

3.8 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustments, and without selection for fit or performance.

3.9 Identification marking. The PUS and major components shall be identified in accordance with MIL-STD-130 and the drawings referenced herein.

3.10 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.6) for manufacturing and testing PUSs and major components when furnished separately:

Item No.	Description	Identification	Quantity
1	Stable base drawings of processor printed wiring board	13220E3120, sheets 2, 3 and 4	1 per sheet (see note 1)
2	Stable base drawings of receiver printed wiring board	13220E3170, sheets 2 and 3	1 per sheet (see note 2)

Notes:
1. Provided only when complete PUSs, processors or processor CCAs are procured.
2. Provided only when complete PUSs, receivers or receiver CCAs are procured.

3.11 Workmanship. Workmanship shall be in accordance with the guidelines and guidance paragraphs of MIL-HDBK-454, guideline 9.

3.11.1 Welding. Sheet metal welding shall be in accordance with AWS D9.1.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 Examination. Each first article PUS and major components when furnished separately shall be examined as specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.2.2 Tests.

4.2.2.1 Complete PUS. The complete PUS shall be subjected to the tests marked "X" in column 1 of table I. Individual testing of major components of the complete PUS is not required. Failure of any test shall be cause for rejection. Unless otherwise specified (see 6.2), five PUSs shall be subjected to environmental tests (see 4.4.2.5), twelve PUSs shall be subjected to the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

reliability test (see 4.4.2.7) and, when required (see 6.2), one PUS, selected from the environmental test units, shall be subjected to the electromagnetic susceptibility test (see 4.4.2.6).

4.2.2.2 Major components furnished separately. Major components furnished separately shall be subjected to the tests marked "X" in column 1 of tables II and III. Failure of any test shall be cause for rejection. Unless otherwise specified (see 6.2), five units shall be subjected to each test.

4.3 Conformance inspection.

4.3.1 Sampling. Sampling for tests shall be in accordance with ANSI/ASQC-Z1.4. Sample size shall be determined by using tables I and IIA, special inspection level S-2. A lot shall be accepted when no defects are found and rejected when one or more defects are found. Samples shall be randomly selected from production lots.

4.3.2 Examination. Each PUS and each major component furnished separately shall be examined for the major defects specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.3.3 Tests.

4.3.3.1 Individual.

4.3.3.1.1 Complete PUS. Each complete PUS shall be subjected to the tests marked "X" in column 2 of table I. Individual testing of major components of the complete UMS is not required. Failure of any test shall be cause for rejection.

4.3.3.1.2 Major components furnished separately. Each major component when furnished separately shall be subjected to the tests marked "X" in column 2 of table II or III, as applicable. Failure of any test shall be cause for rejection.

4.3.3.2 Samples.

4.3.3.2.1 Complete PUS. Each sample, selected in accordance with 4.3.1, shall be subjected to the tests marked "X" in column 3 of table I. Individual testing of major components of the complete PUS is not required. Failure of any test shall be cause for rejection.

4.3.3.2.2 Major components furnished separately. Each sample major component when furnished separately, selected in accordance with 4.3.1, shall be subjected to the tests marked "X" in column 3 of table II or III, as applicable. Failure of any test shall be cause for rejection.

4.4 Inspection procedure.

4.4.1 Examination. The PUS and major components when furnished separately shall be examined as specified herein for the following major defects:

101. Components missing or not as specified (see 3.1, 3.1.1, and 3.1.2).
102. Assembly not correct (see 3.1, 3.1.1, and 3.1.2).
103. Materials not as specified (see 3.3).
104. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).
105. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
106. Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the PUS or major components when furnished separately (see 3.3.3).
107. Interchangeability not as specified (see 3.8).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

108. Identification or marking missing, incomplete or illegible (see 3.9).
109. Workmanship not as specified (see 3.11).
110. Welding not as specified (see 3.11.1).

4.4.2 Tests.

4.4.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at a temperature of 75 ±5 °F and prevailing humidity conditions. Unless otherwise specified herein, the PUS and major components when furnished separately shall operate as specified herein without maintenance.

4.4.2.2 Test schedule.

TABLE I. Test schedule for complete PUS.

First article	Quality conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X X X X	X X - -	- - X X	Functional: (a) Power (b) Performance (c) Detailed circuit (d) Tamper switch tolerance	 3.5.2.1 3.5.2.2 3.5.2.3 3.5.2.4	 4.4.2.4.1 4.4.2.4.2 4.4.2.4.3 4.4.2.4.4
X 1/	-	X	Low temperature	3.4.1(a), 3.4.2(a)	4.4.2.5.1
X 1/	-	X	High temperature	3.4.1(a), 3.4.2(a)	4.4.2.5.2
X 1/	-	X	Humidity	3.4.1(b), 3.4.2(b)	4.4.2.5.3
X 1/	-	-	Shock	3.4.2(c)	4.4.2.5.4
X 1/	-	-	Electromagnetic susceptibility	3.6	4.4.2.6
X 2/	-	-	Reliability	3.7	4.4.2.7

1/ Reliability test units not subjected to this test.
2 Environmental test units not subjected to this test.

TABLE II. Test schedule for receiver and receiver CCA.

First article	Quality conformance		Test 1/	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.5.2.5	4.4.2.4.5
X	-	X	Tamper switch tolerance 2/	3.5.2.4	4.4.2.4.4
X	-	X	Low temperature	3.4.1(a), 3.4.2(a)	4.4.2.5.1
X	-	X	High temperature	3.4.1(a), 3.4.2(a)	4.4.2.5.2
X	-	X	Humidity	3.4.1(b), 3.4.2(b)	4.4.2.5.3
X	-	-	Shock 2/	3.4.2(c)	4.4.2.5.4

1/ The receiver CCA may be placed in a receiver enclosure or equivalent to simulate normal environment.
2/ Required for receiver only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

TABLE III. Test schedule for processor and processor CCA.

First article	Quality conformance		Test 1/	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.5.2.6	4.4.2.4.6
X	-	X	Tamper switch tolerance 2/	3.5.2.4	4.4.2.4.4
X	-	X	Low temperature	3.4.1(a), 3.4.2(a)	4.4.2.5.1
X	-	X	High temperature	3.4.1(a), 3.4.2(a)	4.4.2.5.2
X	-	X	Humidity	3.4.1(b), 3.4.2(b)	4.4.2.5.3
X	-	-	Shock 2/	3.4.2(c)	4.4.2.5.4

1/ The processor CCA may be placed in a processor enclosure or equivalent to simulate normal environment.

2/ Required for processor only.

4.4.2.3 Test methods. Test shall be conducted in accordance with the test methods in Appendix A and as specified herein. Calibration of the test equipment shall be in accordance with ISO 10012-1.

4.4.2.4 Functional.

4.4.2.4.1 Power. The PUS shall be tested in accordance with Appendix A Test Method 1. Nonconformance to 3.5.2.1 shall constitute failure of this test.

4.4.2.4.2 Performance. The PUS shall be tested in accordance with Appendix A Test Method 2. Nonconformance to 3.5.2.2 shall constitute failure of this test.

4.4.2.4.3 Detailed circuit. The PUS shall be tested in accordance with Appendix A Test Method 3. Nonconformance to 3.5.2.3 shall constitute failure of this test.

4.4.2.4.4 Tamper switch tolerance test. The PUS shall be tested in accordance with Appendix A Test Method 4. Nonconformance to 3.5.2.4 shall constitute failure of this test.

4.4.2.4.5 Receiver and receiver CCA. The receiver and the receiver CCA shall be tested in accordance with Appendix A Test Method 6. Nonconformance to 3.5.2.4 and 3.5.2.5 shall constitute failure of this test.

4.4.2.4.6 Processor and processor CCA. The processor and the processor CCA shall be tested in accordance with Appendix A Test Method 7. Nonconformance to 3.5.2.4 and 3.5.2.6 shall constitute failure of this test.

4.4.2.5 Environmental. The PUS and major components when furnished separately shall be subjected to the following environmental tests. The functional test required (see B.3.2.1 of Appendix B) shall be the test specified in 4.4.2.4 applicable to the item being furnished, except for the complete PUS. The functional test for the complete PUS shall be the operational test, Test Method 5 of Appendix A.

4.4.2.5.1 Low temperature. Low temperature test shall be conducted in accordance with low temperature test procedure of Appendix B (see B.4.1). When performing the functional test at -20 °F for the complete PUS (Appendix

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

A, Test Method 5), the amplitude measured at TP2 in step 2.h. shall be within ±3 dB of that measured at room temperature (75 ±5 °F).

4.4.2.5.2 High temperature. High temperature test shall be conducted in accordance with high temperature test procedure of Appendix B (see B.4.2). When performing the functional test at +150 °F for the complete PUS (Appendix A, Test Method 5), the amplitude measured at TP2 in step 2.h. shall be within ±3 dB of that measured at room temperature (75 ±5 °F).

4.4.2.5.3 Humidity. Humidity test shall be conducted in accordance with the humidity test procedure of Appendix B (see B.4.3).

4.4.2.5.4 Shock. Shock test (bench handling) shall be conducted in accordance with shock test of Appendix B (see B.4.4).

4.4.2.5.5 Failure criteria. Nonconformance to 3.4 shall constitute failure of these tests.

4.4.2.6 Electromagnetic susceptibility. When specified (see 6.2), one PUS, consisting of one processor and one receiver, shall be subjected to the following electromagnetic susceptibility tests of MIL-STD-461.

4.4.2.6.1 Radiated susceptibility. Conduct Test Method RS103, except that the electric field requirements of RS103 shall be modified and limited as specified in 3.6.1. All leads (sensor, alarm and power) external to the signal processor enclosure shall be exposed to the electrical field. The leads shall not be less than 4 feet in length.

4.4.2.6.2 Conducted susceptibility. Conduct Test Method CS101, as modified in 3.6.2.

4.4.2.6.3 Failure criteria. Nonconformance to 3.6 shall constitute failure of this test.

4.4.2.7 Reliability. Using an upper test MTBF (θ_0) of 10,000 hours and lower test MTBF (θ_1) of 5,000 hours, the PUS shall be tested as specified herein, with "accept" and "reject" criteria in accordance with MIL-HDBK-781, test plan VIIID. Continue testing until either an "accept" or "reject" decision is reached. All test incidents will be scored by the Government. A failure is defined as any malfunction which causes or may cause one or more of the following:

a. Failure to commence operation, cessation of operation, or degradation of performance below designated levels.
b. Serious damage to the system.
c. Serious personnel safety hazards.

4.4.2.7.1 Test procedure. Unless otherwise specified (see 6.2), twelve complete PUSs, each consisting of one processor and one receiver, shall be randomly selected by a Government representative. The tests shall be conducted under ambient laboratory conditions, without temperature, humidity or vibration cycling. Equipment shall be energized at the power input specified in 3.5.2.1. Operation shall be continuous throughout the test. The 5-second test signal in 2f. of Appendix A test method 5 shall be applied to each PUS not less than nine times during each 24-hour period, but not more often than once per hour. At the end of the reliability test, the PUS shall be subjected to the tests specified in 4.4.2.4.1 and 4.4.2.4.2. The test shall commence only after successful completion of functional tests of each unit. Parts or components of known limited life, such as batteries, shall have a life stipulated prior to test. Failure of these parts occurring after the end of the stipulated period are non-relevant, but any dependent failures caused thereby are relevant. Failures chargeable to workmanship deficiencies shall be counted as relevant. Alarms not deliberately excited as part of the equipment test routine shall be reported and cause of the alarm identified, if

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

practicable. This accounting shall be maintained separately from the failure accounting. Redundant elements, such as indicator lamps, shall not be repaired nor counted as failure until the PUS has failed to perform its intended function. Preventive maintenance procedures specified for the equipment during normal operation shall be performed during the reliability test. No additional preventive maintenance is allowed during this test.

4.4.2.7.2 Failure criteria. Nonconformance to 3.7 shall constitute failure of this test.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES
(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The PUS is intended for use as a penetration detector by receiving and processing the ultrasonic frequency energy generated during forced entry. It is one of the components of the Joint-Services Interior Intrusion Detection System (J-SIIDS).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).

c. Packaging requirements (see 5.1).

d. Nomenclature and quantity of item(s) to be furnished (see 3.1.1).

e. Technical Data Package List, including revision level, to which each item being furnished (by contractor) is to be built.

f. When a first article is required for inspection and approval, the number of units to be subjected to environmental and reliability tests respectively, if other than specified (see 3.2, 4.2.2.1, and 4.2.2.2).

g. When Government-loaned property is not required (see 3.10).

h. When electromagnetic susceptibility testing is required, the number of units to be tested if other than one (see 4.2.2.1 and 4.4.2.6).

6.3 Definitions. The following definitions apply throughout this specification.

6.3.1 False alarm. An alarm generated by stimuli other than those for which the test item is designed to alarm.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

6.3.2 Tamper alarm. Alarm resulting from removal of the sensor assembly cover or pry-off of the sensor assembly from its mounting.

6.3.3 Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, or change of fit of any components; physical change which impairs the mechanical integrity of the PUS; evidence of delamination or water penetration into integrated circuits, printed circuit boards, or external components; or physical condition resulting in noncompliance of the item with the provisions of this specification.

6.3.5 Wideband ultrasonic signal generator. The wideband ultrasonic signal generator (WUSG) produces ultrasonic energy, continuous wave (CW) or pulsed and adjustable in amplitude and in frequency over the range from 10 to 40 kHz. The output signal may be hard-wired to the load or radiated (air-coupled) to the load.

6.3.6 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. A QAP indicates the minimum requirements which must be inspected on the product drawings to verify the objectives of the product and assure interchangeability of the repair parts.

6.4 Quality assurance provisions (QAP) records. The contracting officer should include a clause requiring the contractor to maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. Such clause should also require that the records include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of the end items.

6.6 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.10 when applicable (see 6.2).

6.7 Subject term (key word) listings.

Intrusion detection system
Physical security equipment
Penetration sensor
J-SIIDS

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR4
Navy - SH
Air Force - 99

Preparing activity:
Army - CR4

Project 6350-0208

Review activities:
Army - CR
Navy - CG, MC
DLA - GS

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

TEST METHODS

FOR

PASSIVE ULTRASONIC SENSOR GROUP

A.1. SCOPE

A.1.1 Scope. The test methods contained within this appendix are to determine whether components of the Passive Ultrasonic Sensor Group and major components procured under this specification conform to the requirements specified. Figures 1 and 2 are required to facilitate testing of the group or major components. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

(This section is not applicable to this appendix.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

TEST METHOD 1

POWER TEST

1. TEST APPARATUS:

a. Power supply, dc - constant voltage, adjustable from 18 to 23 Vdc, 100 mA minimum current capacity, line and load regulation 0.5 percent or less, ripple and noise 5 mV P-P.

b. Ammeter, dc - 100 mA minimum, resolution 0.1 mA, accuracy 3 percent, maximum voltage burden 0.3 Vdc.

c. Voltmeter, digital, dc (2 required) - 3 1/2 digit or greater, 10 mV resolution, accuracy ±(0.7 percent of reading ± 1 least significant digit [LSD]).

2. TEST PROCEDURE:

a. Connect 20 receivers to TB2 of the processor by connecting TB1-1 and TB1-2 of each receiver to TB2-1 and TB2-2 respectively of the processor.

b. Place the PUS in operation by applying 20 ±0.3 Vdc through the ammeter to terminals TB1-7 (positive) and TB1-8 of the processor. The current shall not exceed 60 mA.

c. Reverse the polarity on terminals TB1-7 and TB1-8 for 10 seconds. The current shall not exceed 1 mA.

d. Reconnect as in step b. The current shall not exceed 60 mA.

e. Measure the voltage between test point 5 (TP5) (positive) and TB1-8 (common) with a voltmeter. The voltage shall be 12 ±0.6 Vdc.

f. Vary the voltage across TB1-7 and TB1-8 from 18 to 22 V. The voltage at TP5 (12 ±0.6 Vdc) shall not vary more than ±100 mV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

TEST METHOD 2

PERFORMANCE TEST

1. TEST APPARATUS:

a. Power supply, dc (2 required) - constant voltage, adjustable from 20 to 30 Vdc, 100 mA minimum current capacity, line and load regulation 0.5 percent or less, ripple and noise 1 mV P-P.

b. Sound level meter (with microphone) - capable of measuring single frequency radiated ultrasonic energy at frequencies from 10 to 40 kHz, calibrated to measure sound pressure level (SPL) in dB referenced to 0.0002 microbars.

c. Wideband ultrasonic signal generator (WUSG) - capable of radiating ultrasonic acoustic energy, continuous wave (CW) and in bursts of CW for durations of 40 milliseconds (ms) and 5 seconds, at adjustable amplitude from 30 to 80 dB above RSPL over a frequency range from 10 to 40 kHz.

d. Oscilloscope (scope) - general purpose.

e. Signal generator - sine wave of 26.3 kHz, 1 V rms into 600 ohms.

f. AC voltmeter - true rms, calibrated to display the rms value of a waveform and the corresponding dBm (decibels with respect to 1 milliwatt for 600 ohm load), accuracy 1 percent of full scale.

g. Frequency counter - range, 10 to 40 kHz.

2. TEST PROCEDURE:

2.1 Alarm criteria.

a. Place 19 of 20 receivers in an acoustically quiet area. One test receiver shall be excited by the ultrasonic test signals.

b. Connect all 20 receivers to terminal board TB2 of the processor by 500 feet of cable.

c. Set the processor sensitivity to maximum (position 5) and the test receiver gain to 10 dB below maximum (position 7).

d. Apply 30 (+0, -0.3) Vdc through 2.7k ohms across processor terminals TB1-1 and TB1-2 and connect oscilloscope across these terminals to monitor alarm function.

e. Place the PUS in operation by applying 20 ±0.3 Vdc to processor terminals TB1-7 (positive) and TB1-8.

f. Set up the wideband ultrasonic signal generator so that ultrasonic test signals may be generated at any frequency or swept over the range of frequencies between 10 and 40 kHz and at output levels that produce the desired SPL at the receiver microphone. The SPL shall be monitored by placing the sound level meter microphone as close to the receiver as possible and reading SPL at that point.

g. Apply a 5-second burst of ultrasonic test signal at 25.5 kHz and an SPL at the receiver microphone of 60 dB above a reference sound pressure level (RSPL) of 0.0002 microbar. The PUS shall alarm.

h. Apply a 40 ms burst of test signal at 25.5 kHz and 60 dB above the RSPL every 4 seconds. The PUS shall alarm on the fourth burst. As indicated on the oscilloscope, the alarm duration shall be from 0.5 to 1.5 seconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

TEST METHOD 2 - Continued.

2.2 Bandpass characteristics.

a. With the band-reject filter bypassed (jumper between terminals 2 and 3 of TB3 in processor), set the processor sensitivity to midrange (position 3) and the test signal frequency to 25.5 kHz. Adjust the ultrasonic generator SPL to obtain 0.5 V rms at processor TP2 as measured with the ac voltmeter.

b. Vary the test signal frequency from 20 kHz to 30 kHz. The response at TP2 shall not differ by more than 4 dB above to 8 dB below the average in-band response.

c. Keeping the test signal level constant, as in step a., set the test signal frequency to 10 kHz and then to 40 kHz. For both frequencies, the response at TP2 shall be not less than 24 dB below the average in-band response that existed in step b.

2.3 Sensitivity control range.

a. Repeat step 2.2a, but set processor sensitivity to position 5 and adjust ultrasonic generator SPL to obtain 1 V rms at TP2.

b. Set processor sensitivity to minimum (position 1). The signal level at TP2 shall decrease by at least 24 dB.

2.4 Notch filter characteristics.

a. Set the processor sensitivity to position 3 and the test signal frequency to 26.3 kHz. Adjust ultrasonic generator SPL to obtain 0.5 V rms at TP2.

b. Connect the band-reject filter in the processor (remove jumper from terminals 2 and 3 of TB3, and connect jumpers between terminals 1 and 2, and terminals 3 and 4 of TB3).

c. Using a signal generator, apply a sync signal of 26.3 kHz at an amplitude of 1 V rms to TB2-7 in the processor.

d. Keeping the test signal level constant, as in step a, set the test signal frequency to 26.3 +1.315 kHz and then to 26.3 -1.315 kHz. For both frequencies the response at TP2 shall not differ by more than ±3 dB from that measured at 26.3 kHz in step a.

e. Keeping the test signal level constant, set the test signal frequency to 26.3 +0.263 and then to 26.3 -0.263 kHz. For both frequencies the response at TP2 shall be at least 36 dB below that measured at 26.3 kHz in step a.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

TEST METHOD 3

DETAILED CIRCUIT TEST

1. TEST APPARATUS:

a. Power supply, dc (2 required) - constant voltage, adjustable from 18 to 30 Vdc, 100 mA minimum current capacity, line and load regulation 0.5 percent or less, ripple and noise 1 mV P-P.

b. Signal generator - sine wave, 50 ohm output impedance, 25 to 26 kHz minimum frequency range, adjustable output voltage over minimum range 100 mV rms to 1 V rms into 50 ohm load.

c. Step attenuator - calibrated in 10 dB (or less) steps to at least 60 dB, 50 ohm output impedance.

d. AC voltmeter - true rms, calibrated to display the rms value of a waveform, and the corresponding dBm (decibels with respect to 1 miliwatt for 600 ohm load, accuracy 1 percent of full scale.

e. Oscilloscope - general purpose.

f. Voltmeter, dc - digital, 3 1/2 digit or greater, resolution 0.1 Vdc, minimum accuracy ±(0.7 percent of reading ± 1 LSD) or equivalent analog meter.

g. Ammeter, dc - 100 mA minimum, resolution 0.1 mA, accuracy 3 percent, maximum voltage burden 0.3 Vdc.

h. Capacitor - 0.1 microfarad (µf) ±10 percent, 30 working Vdc (WVdc).

2. TEST PROCEDURES:

2.1 Receiver gain control.

a. With TB2-1 and TB2-2 of processor connected to TB1-1 and TB1-2 respectively of receiver and with the band-reject filter in processor bypassed (jumpers disconnected between TB3-1 and TB3-2 and between TB3-3 and TB3-4, and jumper connected from TB3-2 to TB3-3), connect test signal from signal generator through the step attenuator and 0.1 µF capacitor to the junction of R2 and R3 in receiver.

b. Set processor sensitivity to position 3, and receiver gain to maximum (position 10).

c. Place PUS in operation by applying 20 ±0.3 Vdc to processor terminals TB1-7 (positive) and TB1-8.

d. Set test signal frequency to 25.5 kHz and adjust signal generator and attenuator to obtain 1.0 V rms at TP2 of processor, as measured with ac voltmeter.

e. Turn receiver gain control to minimum (position 1). The signal level at TP2 shall decrease by at least 20 dB.

2.2 Alarm duration.

a. With equipment connected as in step 2.1a, apply 30 (+0, -0.3) Vdc through 2.7k ohms to processor terminals TB1-1 and TB1-2 and connect oscilloscope across these terminals.

b. With processor sensitivity set to position 3 and receiver gain set to position 7, adjust 25.5 kHz test signal amplitude to obtain 1 V rms at TP2. Remove test signal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

TEST METHOD 3 - Continued.

c. Apply four 40 ms bursts of test signal within a 15-second period. The oscilloscope shall indicate an alarm on the fourth burst. The alarm duration shall be 1.0 ±0.5 seconds.

2.3 Alarm output.

a. With equipment connected as in step 2.1a, apply 30 (+0, -0.3) Vdc through 2.7k ohms and the ammeter to processor terminals TB1-1 and TB1-2 and connect dc voltmeter across these terminals.

b. Repeat step 2.2b.

c. In the no-alarm state, ammeter shall indicate at least 10 mA, demonstrating 10 mA capability and resistance less than 150 ohms.

d. Momentarily interrupt 20 Vdc power, causing PUS to alarm. Voltameter shall read 30 (+0, -1.2) Vdc, demonstrating 30 Vdc hold and resistance greater than 100k ohms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

TEST METHOD 4

TAMPER SWITCH TOLERANCE TEST

1. TEST APPARATUS:

a. Ohmmeter - Accuracy 3 percent.

b. Test covers - Covers with hole for meter lead access, for processor and receiver.

2. TEST PROCEDURE:

2.1 Processor.

a. Connect an ohmmeter across processor tamper terminals TB1-5 and TB1-6.

b. Place the test cover on the processor. The resistance across the tamper terminals TB1-5 and TB1-6 shall be less than 10 ohms.

c. Raise the cover of the processor more than .062 inch but less than .250 inch (flanges shall be engaged). The resistance between TB1-5 and TB1-6 shall increase to greater than 100K ohms.

d. The resistance between TB1-5 and TB1-6 shall be greater than 100K ohms when the cover is removed.

e. Pull the tamper switch fully out and lock. The resistance between TB1-5 and TB1-6 shall decrease to less than 10 ohms.

2.2 Receiver.

a. Connect an ohmmeter across receiver tamper terminals TB1-5 and TB1-6.

b. Place the test cover on the receiver. The resistance across the tamper terminals TB1-5 and TB1-6 shall be less than 10 ohms.

c. Raise the cover on the receiver more than .062 inch but less than .250 inch (flanges shall be engaged). The resistance across TB1-5 and TB1-6 shall increase to greater than 100k ohms.

d. The resistance between TB1-5 and TB1-6 shall be greater than 100k ohms when the cover is removed.

e. Pull the tamper switch fully out and lock. The resistance between TB1-5 and TB1-6 shall decrease to less than 10 ohms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

TEST METHOD 5

OPERATIONAL TEST

1. TEST APPARATUS:

a. Power supply, dc (2 required) - constant voltage, adjustable 0 to 30 Vdc, 100 mA minimum current capacity, line and load regulation 0.05 percent or less, maximum ripple and noise 5 mV P-P.

b. Oscilloscope - general purpose.

c. Signal generator (2 required) - sine wave, 50 ohm output impedance, 10 to 40 kHz minimum frequency range at constant voltage output, output voltage adjustable over minimum range of 1 to 10 V P-P.

d. Step attenuator - calibrated in 10 dB steps to at least 80 dB, 50 ohm output and input impedances.

e. AC voltmeter - calibrated in volts rms and in dBm for 600 ohm load, range 0 to -100 dBm minimum, accuracy 1 dBm.

f. Voltmeter, dc - digital, 3 1/2 digit or greater, resolution 0.1 Vdc, minimum accuracy 1.0 percent (or equivalent analog meter).

g. Ohmmeter - multiple range, minimum resolution 1 percent of reading at 150 ohms and 100k ohms, accuracy 3 percent.

h. Tone burst generator - Gated and timing voltage inputs and outputs, on and off cycle counts as necessary to generate 5 seconds continuous signal and bursts of 40 ms on, 4 seconds off, when used with signal generator.

i. Frequency counter - range 20 to 30 kHz minimum.

j. Capacitor - 0.1 µf, ±5 percent, 50 WVdc.

2. TEST PROCEDURE:

a. With TB2-1 and TB2-2 of processor connected to TB1-1 and TB1-2 respectively of receiver, and with band-reject filter connected in the processor, connect test signal from the signal generator through the step attenuator and the 0.1 µF capacitor to the junction of R2 and R3 in receiver. From the second signal generator, apply a 26.3 kHz sync signal at 1 V rms to TB2-7 in the processor.

b. Set processor sensitivity to position 3 and receiver gain to position 7.

c. Apply 30 (+0, -0.3) Vdc through 2.7k ohms to processor terminals TB1-1 and TB1-2 and connect the oscilloscope across these terminals.

d. Place PUS in operation by applying 20 ±0.3 Vdc across processor terminals TB1-7 (positive) and TB1-8.

e. Set test signal to 25.5 kHz and adjust input level to obtain 1.2 V rms at processor TP2. Alarm pulses on scope shall occur every 4 to 6 seconds.

f. Remove test signal and momentarily short TP4 in processor to ground. Then apply a 5 second test signal at 25.5 kHz. The PUS shall alarm in 4 to 5 seconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

TEST METHOD 5 - Continued.

g. Momentarily short TP4 to ground. Apply 40 ms test signals at 4 to 5 second intervals. The PUS shall alarm on the fourth test signal. The alarm duration observed on the oscilloscope shall be 1.0 ±0.5 seconds.

h. Vary the test signal frequency from 20 to 30 kHz. The signal amplitude at TP2 shall be 1.2 V rms ±3 dB for signals from 20 to 24.985 kHz and 27.615 to 30 kHz.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

TEST METHOD 6

RECEIVER AND RECEIVER CIRCUIT CARD ASSEMBLY

(TA13220E3151 and TA13220E3163)

1. TEST APPARATUS:

a. Power supply, dc - constant voltage, adjustable from 19 to 21 Vdc, 10 mA minimum current capacity, line and load regulation 0.05 percent or less, ripple and noise 1 mV P-P.

b. Oscilloscope - 10 MHz bandwidth, vertical deflection factors of 5, 10, 20 and 50 mV per div., 6 vertical divisions minimum, selectable horizontal sweep rates from 5 microseconds per div. to 100 microseconds/div., 8 div. minimum, internal/external trigger, adjustable trigger level, ac and dc coupled inputs, vertical/horizontal deflection accuracy 3 percent.

c. Signal generator - sine wave, 50 ohm output impedance, 5 kHz to 50 kHz minimum frequency range, amplitude change with frequency less than 0.2 dB over 5 to 50 kHz range, adjustable output voltage over minimum range 100 mV rms to 1 V rms into 50 ohm load, frequency drift no more than 0.05 percent of setting in 10 minutes, amplitude stability ±0.05 percent of maximum P-P values in 10 minutes.

d. Step attenuator - calibrated in 10 dB (or less) steps to at least 60 dB, 50 ohm output impedance.

e. Voltmeter, dc (M1, M2) - digital, 3 1/2 digit or greater, resolution 0.1 Vdc, minimum accuracy 0.5 percent of reading ±2 LSD or equivalent analog meter.

f. AC voltmeter - calibrated in volts rms and in dBm for 600 ohm load, range 0 to -100 dBm minimum, and corresponding voltage ranges, accuracy 3 percent of full scale at 50 Hz to 50 kHz.

g. Ohmmeter, (M3) - multiple range, minimum resolution 1 percent of reading at 10 ohms and 100k ohms, accuracy 3 percent.

h. Sound pressure meter, consisting of: following, or equivalent:
(1) B&K measuring amplifier, model 2606;
(2) B&K preamplifier, model 2618;
(3) B&K microphone model 4135.

i. Wideband ultrasonic signal generator - radiates continuous wave acoustic ultrasonic energy, amplitude adjustable to 75 dB above RSPL minimum at one foot on axis, at frequencies from 20 to 30 kHz.

j. Resistor - 18k ohms, ±5 percent, 0.1 watt or greater.

k. Resistor - 24k ohms, ±5 percent, 0.1 watt or greater.

l. Resistor - 50 or 51 ohms, ±10 percent, 1/4 watt.

m. Capacitor (2 required) - 0.1 µf, ±10 percent, 30 WVdc.

n. Test fixture - (see figure 1).

2. TEST PROCEDURE:

2.1 General.

a. Prior to mounting the microphone in the circuit card assembly, acceptance tests 2.2 through 2.5 shall be completed. For first article

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

receivers and receivers selected for sample tests, an appropriate socket may be fabricated or purchased to facilitate repeated installation and removal, as long as the microphone is soldered directly to the circuit card for the final test.

b. Connect the circuit card assembly to the test fixture (figure 1).

2.2 Input power.

a. Place the receiver in operation by applying 20 ±0.3 Vdc as indicated on meter M1. The dc voltage across TB1-1 and TB1-2 of the circuit card shall be 5 ±0.5 Vdc as indicated on meter M2.

2.3 Frequency response.

a. Set the signal generator frequency to 25 kHz and the step attenuator, if required (see note 1 on figure 1), to 50 dB.

b. Set the receiver gain control to position 5. Adjust the signal generator to obtain a 50 mV P-P sine wave at TP-1 on the test fixture, as indicated on the oscilloscope.

c. Vary the signal generator over the 20 to 30 kHz frequency range and observe the output signal on the oscilloscope.

d. The output signal at TP-1 shall not vary more than ±5 mV from the initial 50 mV P-P signal.

2.4 Receiver gain control.

a. Set the receiver gain control to position 10 and set the signal generator to 22 kHz.

b. Adjust the signal generator for a 150 mV P-P signal at TP-1 as indicated on the oscilloscope.

c. Set the receiver gain control to position 1 and observe the signal on the oscilloscope.

d. The signal at TP-1 shall be not greater than 15 mV P-P (down at least 20 dB).

2.5 Tamper switch.

a. Connect an ohmmeter (M3) to the tamper terminals TB1-5 and TB1-6 of the receiver card.

b. Depress the tamper switch. The resistance across TB1-5 and TB1-6 shall be less than 10 ohms.

c. Release the tamper switch. The resistance across the tamper terminals shall be greater than 100k ohms.

d. Pull the tamper switch fully out and lock. The resistance across TB1-5 and TB1-6 shall be less than 10 ohms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

TEST METHOD 6 - Continued.

2.6 Sound pressure level.

Note: During environmental tests, when performing the sound pressure level measurements, the receiver or receiver CCA shall be removed from the environmental chamber to preclude reflections. Measurements shall be performed within 5 minutes after removal from the chamber (45 minutes for humidity tests).

a. Turn off the 20 Vdc power supply and disconnect the signal generator, attenuator, and test leads from the receiver test fixture (figure 1). Substitute the 18k ohm resistor for the 24k ohm resistor in the test fixture.

b. Mount the microphone in the receiver circuit card assembly and set the gain control to position 5.

c. Place the receiver circuit card in a vertical position with the microphone in the direction of the ultrasonic signal generator transmitting transducer at a distance of 1 foot.

Note: The ultrasonic test signal is applied by operating the wideband ultrasonic generator at the desired frequency and at an output level that produces the desired SPL at the receiver microphone. This is monitored by placing the sound level meter microphone as close to the receiver microphone as possible and reading the SPL at that point.

d. Place the receiver circuit card assembly in operation by applying 20 ±0.3 Vdc as indicated on meter M1. The dc voltage across terminals TB1-1 and TB1-2 of the card shall be 5 ±0.5 Vdc as indicated on meter M2.

e. Adjust the wideband ultrasonic generator to provide a 75 dB above RSPL signal at 25 kHz as measured with the sound pressure level meter microphone.

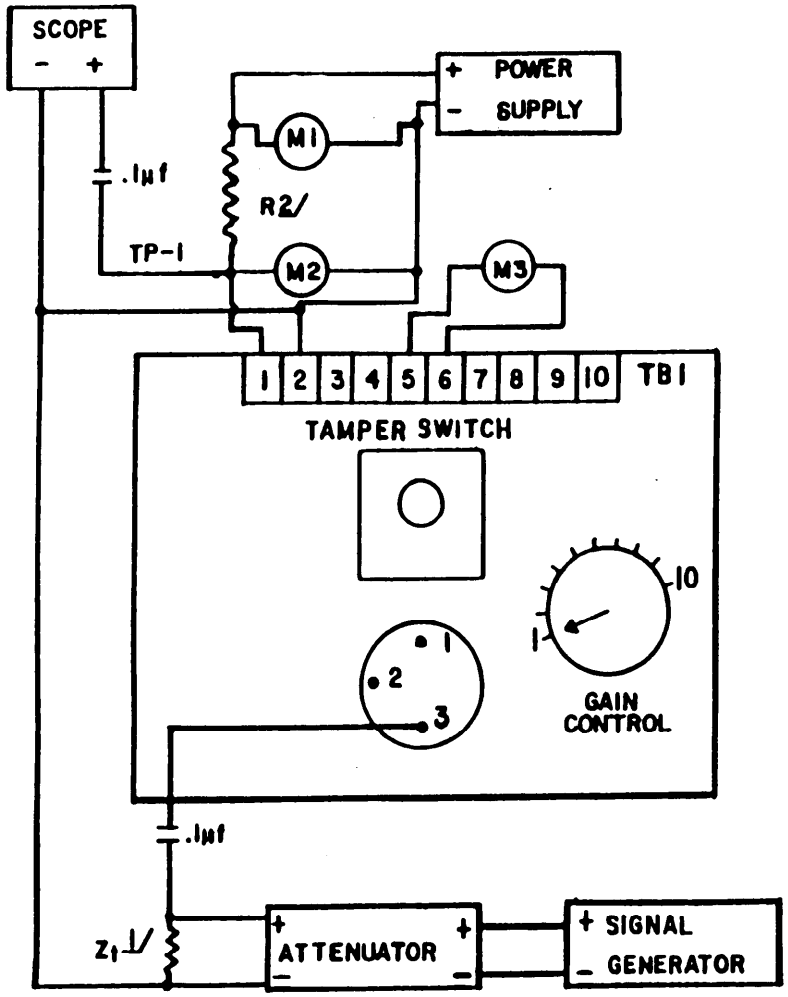
f. Measure the ac signal at TP-1 with the scope. The signal shall be 20 ±8 mV rms for acceptance. A voltmeter may be used instead of the scope to measure voltage as long as it has been verified that no serious distortion of the waveform exists. Up to 10 percent jitter is acceptable.

g. Repeat steps e. and f. with the wideband ultrasonic generator adjusted for 20 and 30 kHz.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

TEST METHOD 6 - Continued.

FIGURE 1. PUS RECEIVER TEST FIXTURE. X-3925A



NOTES:

1/ Z1 = 51 OHMS FOR ATTENUATOR HAVING 50 OHM OUTPUT IMPEDANCE. WHEN USING A SIGNAL GENERATOR THAT IS CAPABLE OF PRODUCING AN UNDISTORTED SIGNAL AT VOLTAGES AS LOW AS 0.8 mV RMS, THE Z1 AND ATTENUATOR MAY BE OMITTED.

2/ R = 24K OHMS ±5 PERCENT FOR TESTS WITH MICROPHONE OUT AND 18K OHMS ±5 PERCENT FOR TESTS WITH MICROPHONE INSTALLED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

TEST METHOD 7

PROCESSOR AND PROCESSOR CIRCUIT CARD ASSEMBLY

(TA13220E3101 AND TA13220E3111)

1. Test Apparatus.

a. Power supply, dc (2 required) - constant voltage, adjustable 0 to 30 Vdc, 100 mA minimum current capacity, line and load regulation 0.05 percent or less, maximum ripple and noise 5 mV P-P.

b. Oscilloscope (2 required) - 20 MHz minimum bandwidth, deflection factors of 10 mV or less to 5 V or more per division, 8 vertical divisions minimum, selectable horizontal sweep rates from less than or equal to 1 microsecond per div. to greater than or equal to 2 seconds/div., internal/external trigger, adjustable trigger level, ac and dc coupled inputs.

c. Signal generator (3 required) - 50 ohm output impedance, 10 to 40 kHz minimum frequency range, constant voltage output over frequency range, adjustable output voltage, minimum range 1 V P-P to 10 V P-P.

d. Step attenuator - calibrated in 10 dB steps to at least 80 dB, 50 ohm output and input impedances.

e. Step attenuator - calibrated in 1 dB steps to at least 10 dB, 50 ohm output and input impedances.

f. Voltmeter, dc (3 required) - digital, 3 1/2 digit or greater, resolution 0.1 Vdc, minimum accuracy 0.5 percent of reading ±2 LSD (or equivalent analog meter).

g. Ohmmeter (M1) - multiple range, minimum resolution 3 percent of readings of 10 ohms, 150 ohms and 100k ohms, accuracy 3 percent.

h. Tone burst generator - Gated and timing voltage inputs, on and off cycle counts as necessary to generate 5 seconds on and bursts of 40 ms on, 4 seconds off, when used with signal generator.

i. Frequency counter - range 10 to 40 kHz minimum, resolution 1 Hz.

j. AC voltmeter (2 required) - calibrated in volts rms and in dBm for 600 ohm load, range 0 to -100 dBm minimum, accuracy 3 percent.

k. Resistor (R1) - 600 ohms, 1 percent, 1/4 watt.

l. Resistor (R2) - 2700 ohms, 5 percent, 1/2 watt.

l. Capacitor (C1) - 0.1 µF, ±5 percent, 50 WVdc.

m. Test fixture - (see figure 2).

2. Test Procedure.

2.1 General.

a. Disconnect jumpers between terminals 1 and 2 and terminals 3 and 4 of TB3.

b. Connect jumper between terminals 2 and 3 of TB3.

c. Connect the circuit card assembly in the test fixture (figure 2).

d. Do not connect signal generator number 2 until required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

TEST METHOD 7 - Continued.

e. All test signals required below shall be sine waves.

2.2 Input power.

a. Place the processor in operation by applying 20 ±0.3 Vdc through the ammeter to terminals TB1-7 (positive) and TB1-8 of the processor. The current shall not exceed 35 mA.

b. Reverse the polarity on terminals TB1-7 and TB1-8 for 10 seconds. The current shall not exceed 1 mA.

c. Reconnect as in step a. The current shall not exceed 35 mA.

d. Measure the voltage at TP5 with a voltmeter. The voltage shall be 12 ±0.6 Vdc.

e. Vary the voltage at terminals TB1-7 and TB1-8 from 18 to 22 V. The voltage at TP5 shall not vary more than ±100 mV.

2.3 Alarm criteria.

a. Set the processor sensitivity to maximum (position 5).

b. Apply 30 (+0, -0.3) Vdc through 2.7k ohms to terminals TB1-1 and TB1-2 and connect oscilloscope across these terminals to monitor alarm function.

c. Place the processor in operation by applying 20 ±0.3 Vdc to terminals TB1-7 and TB1-8.

d. Apply a 5-second burst of test signal at frequency of 25.5 kHz from signal generator number 1 at an amplitude of -75 ±3 dBm across TB2-1 and TB2-2. The processor shall alarm.

e. Apply 40 ms bursts of the 25.5 kHz test signal at an amplitude of -75 ±3 dBm across TB1 and TB2 every 4 seconds. The processor shall alarm on the fourth burst. As indicated on the oscilloscope, the alarm duration shall be from 0.5 to 1.5 seconds.

2.4 Bandpass characteristics.

a. With the band-reject filter bypassed (jumper installed between terminals 2 and 3 of TB3), set the processor sensitivity to mid-range (position 3) and the test signal frequency to 22 kHz. Adjust signal generator number 1 and the attenuator so as to obtain 0.5 V rms at TP2 as measured with the ac voltmeter. The attenuator must be set to at least 50 dB. Record the attenuator setting. Do not change the signal generator amplitude during the remaining steps of 2.4.

b. Vary the test signal frequency from 20 to 30 kHz. The response at TP2 shall not differ by more than ±3 dB from the average in-band response.

c. Keeping the signal generator number 1 test signal amplitude constant, set the test signal frequency to 10 kHz and adjust the attenuator to obtain 0.5 V rms at TP2. Record the attenuator setting. This reading shall be at least 35 dB less than the reading recorded in step a.

d. Repeat step c. using 40 kHz in place of 10 kHz. The attenuator setting shall be at least 20 dB less than the reading recorded in step a.

2.5 Sensitivity control range.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

TEST METHOD 7 - Continued.

a. Repeat step 2.4a, but set processor sensitivity to position 5, and adjust signal generator number 1 and attenuator to obtain 1 V rms at TP2. The attenuator must be set to at least 50 dB. Record the attenuator setting.

b. Set processor sensitivity to minimum (position 1). Adjust the attenuator to obtain 1 V rms at TP2 and record the attenuator setting. This setting shall be at least 24 dB less than the reading recorded in step a.

2.6 Notch filter characteristics.

a. Set the processor sensitivity to position 3 and the frequency to 26.3 kHz. Adjust signal generator number 1 and attenuator to obtain 0.5 V rms at TP2. The attenuator must be set to at least 50 dB. Record the attenuator setting.

b. Connect the band-reject filter in the processor (remove jumper from terminals 2 and 3 of TB3) and connect jumpers between terminals 1 and 2 and terminals 3 and 4 of TB3.

c. Using signal generator number 2, apply a sync signal of 26.300 ±0.020 kHz at 1 V rms to TB2-7 in the processor.

d. Keeping the test signal level from the signal generator number 1 constant, as in step a., set the test signal frequency to 26.300 +1.315 kHz and then to 26.300 -1.315 kHz. For both frequencies, the response at TP2 shall not differ by more than ±3 dB from that measured at 26.3 kHz in step a.

e. Keeping the test signal amplitude from the signal generator constant, set the test signal frequency to 26.300 +0.263 kHz, and then to 26.300 -0.263 kHz. For both frequencies adjust the attenuator to obtain 0.5 V rms at TP2 and record attenuator reading for both. The readings for both shall be at least 36 dB less than the reading recorded in step a.

2.7 Alarm output.

a. Apply 30 (+0, -0.3) Vdc through 2.7k ohms and the ammeter to processor terminals TB1-1 and TB1-2 and connect dc voltmeter across these terminals.

b. In no-alarm state, the ammeter shall indicate at least 10 mA, demonstrating 10 mA capability and resistance less than 150 ohms.

c. Momentarily interrupt 20 Vdc power, causing processor to alarm. Voltmeter shall read 30 (+0, -1.2) Vdc, demonstrating 30 Vdc hold and resistance greater than 100k ohms.

2.8 Tamper switch.

a. Connect an ohmmeter (M1) to tamper terminals TB1-5 and TB1-6 of the processor card.

b. Depress the tamper switch. The resistance across tamper terminals TB1-5 and TB1-6 shall be less than 10 ohms.

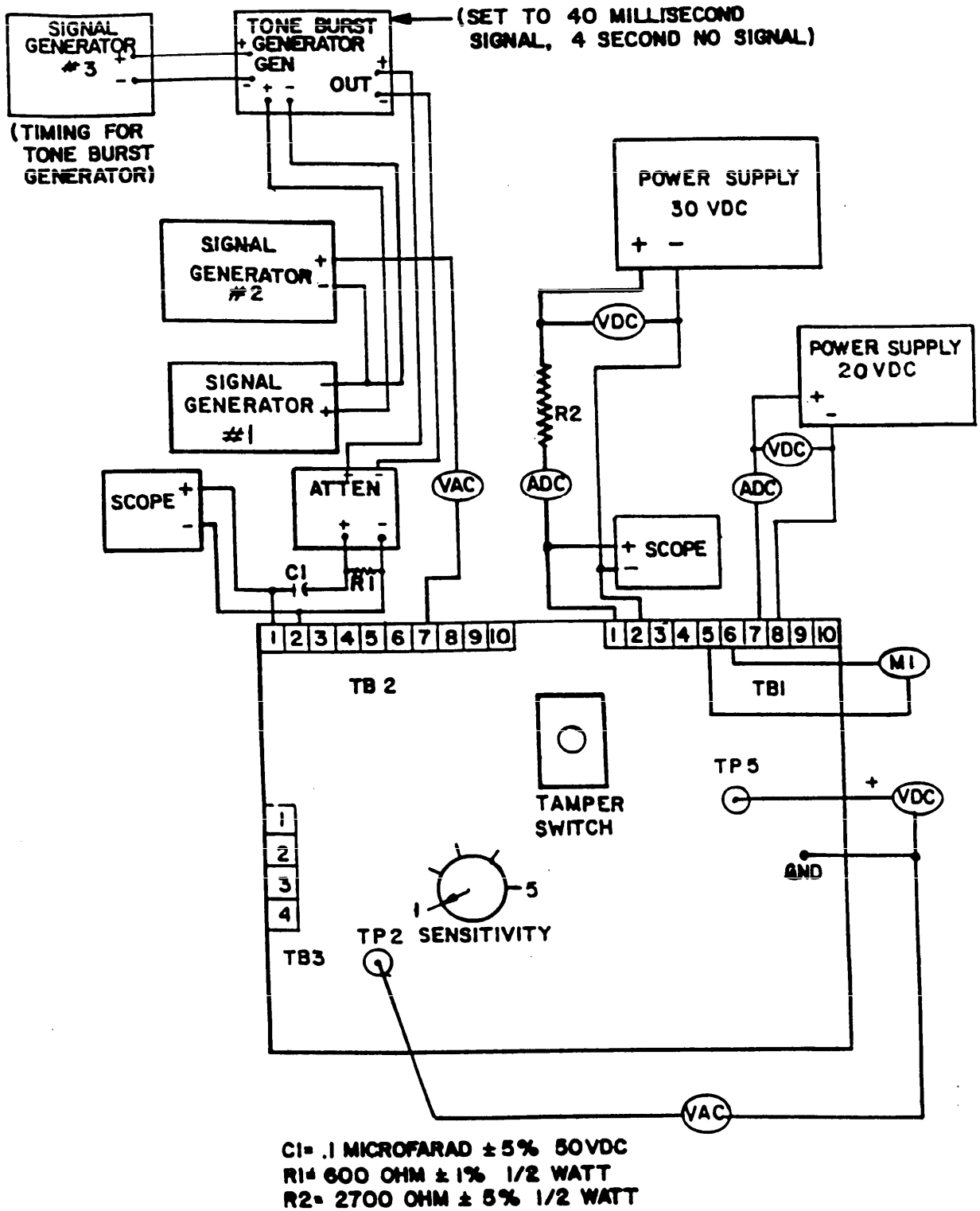
c. Release the tamper switch. The resistance across the tamper terminals shall be greater than 100k ohms.

d. Pull the tamper switch fully out and lock. The resistance between terminals TB1-5 and TB1-6 shall be less than 10 ohms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

TEST METHOD 7 - Continued.

FIGURE 2. PUS PROCESSOR TEST FIXTURE. X-3926A



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

ENVIRONMENTAL TEST METHODS

B.1 SCOPE

B.1.1 Scope. The environmental test procedures contained in this appendix are to determine whether PUSs and major components furnished separately procured under this specification conform to the environmental requirements specified. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

B.2 APPLICABLE DOCUMENTS

(This section is not applicable to this appendix.)

B.3 GENERAL REQUIREMENTS

B.3.1 Test conditions. Unless otherwise specified herein, tests shall be performed under the following standard ambient conditions:

Temperature: 75 ±5 °F
Relative humidity: 50 ±30 percent.

B.3.1.1 Measurements of test conditions. All measurements of test conditions shall be made with instruments of the accuracy specified in B.3.1.3.

B.3.1.2 Tolerance of test conditions. Unless otherwise specified, tolerance of test conditions shall be as follows:

a. Air temperature at the control sensor: ±2.5 °F. The equipment sensor response time (T) shall be 20 seconds or less. Temperature gradient across the cross-sectional area occupied by the test item shall not exceed 0.5 °F per foot in any direction, but never more than 4 °F total (equipment nonoperating).

b. Relative humidity at the control sensor: ±5 percent.

NOTE: (T) is the time required for the sensing system to respond to 62.3 percent of a step change in temperature in the measured environment.

B.3.1.3 Accuracy of test apparatus. The accuracy of instruments and test equipment used to control or monitor the test parameters shall be verified and shall satisfy the requirements of ISO 10012-1 to the satisfaction of the procuring activity. All instruments and test equipment used in conducting the tests specified herein shall:

a. Conform to laboratory standards whose calibration is traceable to the prime standards at the US Bureau of Standards.

b. Have an accuracy of at least one-third the tolerance for the variable to be measured. In the event of conflict between this accuracy and a requirement for accuracy in any one of the test methods of this standard, the latter shall govern.

B.3.1.4 Stabilization of test temperature.

B.3.1.4.1 Operating. Unless otherwise specified, temperature stabilization will have been attained when the temperature of the part of the test item considered to have the longest thermal lag is changing no more than 3.6 °F per hour.

B.3.1.4.2 Non-operating. Unless otherwise specified, temperature stabilization will have been attained when the temperature of the part of the test item considered to have the longest thermal lag reaches a temperature

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

within 3.6 °F of the prescribed temperature. When changing temperatures, the temperature of the chamber air may be adjusted up to 9 °F beyond the desired end point for a period of time of up to 1 hour to reduce stabilization time, provided that the stabilization requirements of this paragraph are ultimately attained relative to the specified end point temperature, and provided the extended chamber temperatures will not cause damage to the test item.

B.3.2 Performance of test.

B.3.2.1 Functional test. The functional test referenced in these environmental test procedures shall be that specified in 4.4.2.5.

B.3.2.2 Pre-test functional test. Prior to proceeding with each environmental test, the test item shall be subjected to the applicable functional test (see B.3.2.1) under standard ambient conditions (see B.3.1). After the first environmental test (low temperature), this may be the post-test functional test (see B.3.2.5) after the previous environmental test.

B.3.2.3 Installation of test item in test chamber. The test chamber shall be at standard ambient conditions when the test item is installed. Suitable test leads shall be attached to each test item and brought out through port hole(s) to allow complete functional tests without opening the door of the test chamber during high temperature and low temperature tests. The test item shall then be subjected to the applicable functional test or an abbreviated functional test sufficient to determine that no malfunction or damage was caused due to faulty installation or handling.

B.3.2.4 Functional test at operating temperature. Perform applicable functional test (see B.3.2.1) during the test exposure as specified in the test procedure herein.

B.3.2.5 Post-test functional test. At the completion of each environmental test, the test item shall be subjected to the applicable functional test (see B.3.2.1) under standard ambient conditions (see B.3.1).

B.3.2.6 Failure criteria.

a. The item shall have failed the test when any of the following occur:

(1) Functional test results not in accordance with applicable test method during or after any of the tests.

(2) Deterioration, corrosion, damage or change in tolerance limits of any internal or external parts which could in any manner prevent the test item from meeting operational, service or maintenance requirements.

B.3.3 Test facilities and apparatus. Test facilities, chambers, and apparatus used in conducting the tests contained in this appendix shall be capable of meeting the conditions required.

B.3.3.1 Test chamber.

B.3.3.1.1 Volume of test chamber. The volume of the test chamber shall be such that the bulk of the item under test will not interfere with the generation and maintenance of the test conditions. When testing multiple sample items simultaneously, the test chamber shall be of sufficient size so that each test unit is provided uniform environmental conditions and is not subjected to non-test environments.

B.3.3.1.2 Heat source. The heat source of the test facility shall be so located that radiant heat from the source will not fall directly on the test item.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

B.3.3.1.3 Location of temperature sensors. Unless otherwise specified, thermocouples or equivalent temperature sensors utilized to determine or control the specified chamber temperature shall be located centrally within the chamber, in the supply airstream, or in the return airstream, whichever provides the specified test conditions at the bulk under test and shall be baffled or otherwise protected against radiation effects.

B.3.3.1.4 Internal air circulation. The conditioned air flow shall be suitably baffled to provide uniform airflow around the test item. If multiple test items are tested, they shall be so spaced as to provide free circulation between the test items and the chamber walls.

B.3.4 Test data. Test data shall include complete identification of all test equipment and accessories. The data shall include the actual test sequence used and ambient test conditions recorded periodically during the test period. The test record shall contain a signature and date block for certification of the test data by the test engineer.

B.3.5 Test sequence. Units being subjected to environmental tests shall have been functionally tested at room ambient temperature prior to environmental tests. Environmental tests shall be performed in the sequence low temperature, high temperature, humidity and shock, unless otherwise approved by procuring contracting officer.

B.4 DETAILED TEST PROCEDURES

B.4.1 Low temperature test procedure.

B.4.1.1 Apparatus. Temperature chamber.

B.4.1.2 Procedure.

a. Prepare the test item(s) in accordance with General Requirements (see B.3).

b. Lower the internal chamber temperature to the storage temperature, -30 °F, and maintain for a period of 48 hours after stabilization.

c. Adjust the internal chamber temperature to -20 °F and maintain until temperature stabilization of the test item is reached or for one hour after the chamber temperature sensor has reached -20 °F.

d. Apply operating power to the test item until the item is stabilized.

e. Perform the functional test of B.3.2.1 applicable to the test item.

f. Return the test item, non-operating, to standard ambient conditions and maintain for 2 hours or until temperature stabilization of the test item is reached.

g. Apply operating power, allow test item to stabilize, and perform the functional test conducted in step e. above (post-low temperature test).

NOTE: The rate of temperature change during steps b., c., and f. may be the maximum attainable by the chamber but shall not exceed 18 °F per minute.

B.4.2 High temperature test procedure.

B.4.2.1 Apparatus. Temperature chamber.

B.4.2.2 Procedure.

a. Prepare the test item in accordance with General Requirements (see B.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

b. Raise the internal chamber temperature to 120 °F.

c. Maintain internal chamber temperature for 6 hours at 120 °F.

d. Raise the internal chamber temperature to 165 °F within a time period of l hour and then maintain at that temperature for 4 additional hours.

e. Lower the internal chamber temperature to 120 °F within a time period of 1 hour.

f. Repeat steps c., d., and e. two additional times (making a total of three 12-hour cycles).

g. Adjust the internal chamber temperature to +150 °F and maintain for 2 hours or until temperature stabilization of the test item is reached.

h. Apply power to the test item until the test item is stabilized and perform the functional test of B.3.2.1 applicable to the test item.

i. Return the test item, non-operating, to standard ambient conditions and maintain for 2 hours or until temperature stabilization of the test item is reached.

j. Apply power to the test item until the item is stabilized and perform the test conducted in step h. above (post-high temperature test).

B.4.3 Humidity test procedure.

B.4.3.1 Apparatus. Humidity-temperature chamber and associated equipment.

B.4.3.1.1 Chamber. The chamber and accessories shall be constructed and arranged in such a manner as to avoid condensate dripping on the test item. The chamber shall be trap-vented to the atmosphere to prevent the buildup of total pressure. Relative humidity shall be determined from the dry bulb-wet bulb thermometer comparison method or an equivalent method approved by the procuring activity. When readout charts are used, they shall be capable of being read with a resolution within ±1 °F. When the wet bulb control method is used, the wet bulb and tank shall be cleaned and a new wick installed at least every 30 days. The air velocity flowing across the wet bulb shall be not less than 900 feet per minute. Provisions shall be made for controlling the flow of air throughout the internal chamber test space where the velocity of air shall not exceed 150 feet per minute. Steam or distilled, demineralized, or de-ionized water having a pH value between 6.0 and 7.2 at 73 °F shall be used to obtain the specified humidity. No rust or corrosive contaminants shall be imposed on the test item by the test facility.

B.4.3.1.2 Procedure.

a. Prepare the test item in accordance with General Requirements (see B.3). Prior to starting the test the internal chamber temperature shall be at standard ambient temperature (see B.3.1) with uncontrolled humidity.

b. Gradually raise internal chamber temperature to 149 °F and the relative humidity to 95 percent or greater over a period of 2 hours.

c. Maintain conditions of step b for not less than 6 hours.

d. Maintain 92 percent, or greater, relative humidity and reduce internal chamber temperature in 16 hours to 86 °F. Seventy five minutes before the end of the 16-hour period, commence performing a complete functional test of the test items in accordance with B.3.2.1. The test items may be removed from the environmental chamber for this test; however, unsuccessful completion of test or unsuccessful return of the test items to

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

the test chamber within 75 minutes after initiation of tests shall constitute failure of the test. Failure to conform to the parameters specified in the functional test procedure shall also constitute failure of the test.

e. Repeat steps b., c., and d. for a total of 5 cycles (not less than 120 hours). Perform functional testing, as in step d. above, commencing 75 minutes before the completion of each 24-hour cycle. Figure 3 is an outline of the humidity cycle for this procedure.

f. After the completion of the fifth test period, return test items to standard ambient conditions, allow test item to reach temperature stabilization and perform functional test as in step d.

NOTE: The equipment shall be operated only when required tests are being performed.

B.4.4 Shock test (bench handling). This procedure shall be used to determine the ability of equipment to withstand the shock encountered during servicing. Remove the cover to perform test. The test item shall be placed in a suitable position for servicing on a horizontal, solid wooden bench top at least 1-5/8 inches thick. The test shall be performed, as follows, in a manner simulating shocks likely to occur during servicing.

Step 1. Using one edge as a pivot, lift the opposite edge of the chassis until one of the following conditions occurs (whichever occurs first):

a. The chassis forms an angle of 45 degrees with the horizontal bench top.

b. The lifted edge of the chassis has been raised 4 inches above the horizontal bench top.

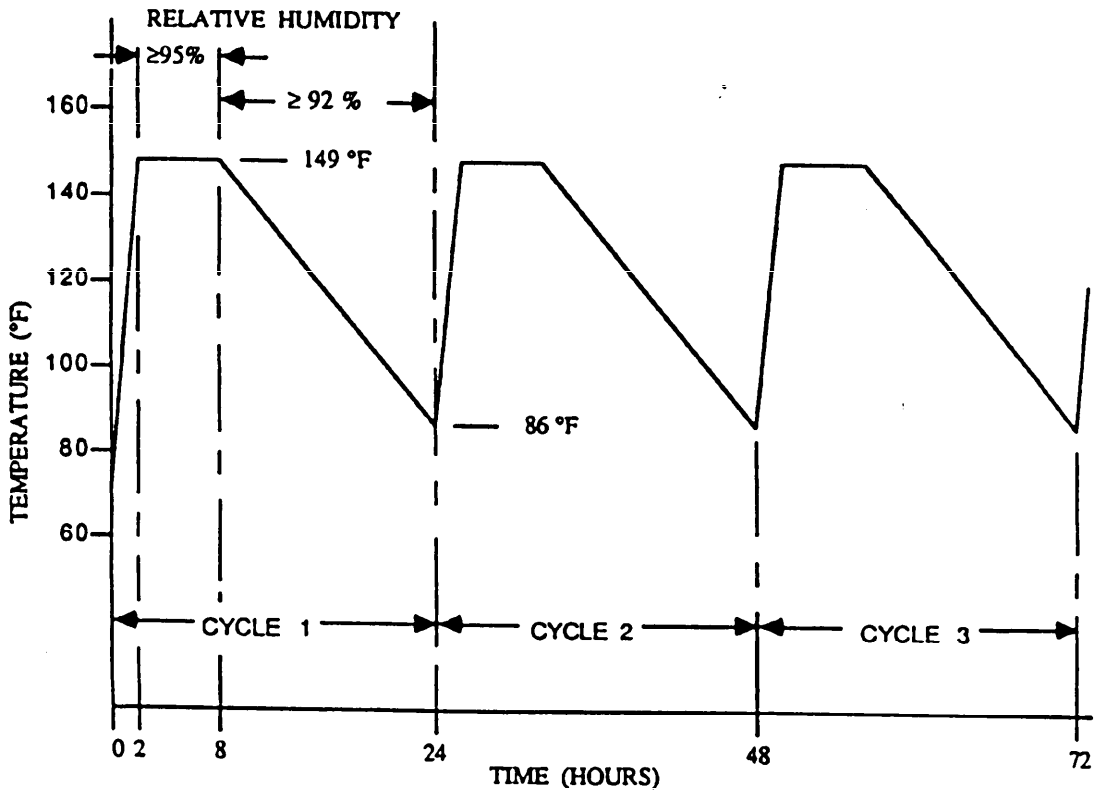
c. The lifted edge of the chassis is just below the point of perfect balance.

Let the chassis drop back freely to the horizontal bench top. Repeat, using other practical edges of the same horizontal face as pivot points, for a total of four drops.

Step 2. Repeat step 1, with the test item resting on other faces until it has been dropped for a total of four times on each face on which the test item could be placed practicably during servicing. The test item shall not be operating during the test. At the conclusion of the test, the test item shall be inspected and subjected to the applicable functional test of B.3.2.1. Failure criteria shall be as specified in B.3.2.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

FIGURE 3. TEST CYCLE FOR HUMIDITY TEST. X-5056



NOTE: CONTINUE FOR A TOTAL OF 5 CYCLES (120 HOURS)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter must be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-52869C	2. DOCUMENT DATE (YYMMDD) 970630

3. DOCUMENT TITLE
Sensor Group, Passive Ultrasonic: Receiver R-1860/FSS-9(V) and Processor MX-9443/FSS-9(V)

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)***

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME BELVOIR CECOM SOUTH	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468 (2) AUTOVON DSN 654-3468
c. ADDRESS (Include Zip Code) BELVOIR CECOM SOUTH 10115 GRIDLEY ROAD STE 228 ATTN: AMSEL-LC-IEW-D-ED Fort Belvoir VA 22060-5849	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.